EXERCISES WEEK 7 – ANSWERS

8CM10 - Introduction to Quantum Physics and Structure of Matter

NUCLEAR MASSES AND BINDING ENERGIES

1. Compute the binding energy B_{tot} and the binding energy per nucleon B_{tot}/A for:

   a) $^{208}\mathrm{Pb}$
   b) $^{133}\mathrm{Cs}$
   c) $^{90}\mathrm{Zr}$
   d) $^{59}\mathrm{Co}$

   For the atomic masses, check the internet or the tables at the end in the book of Krane.

(a) $B = [Nm_{\mathrm{n}} + Zm(^{1}\mathrm{H}) - m(^{A}X)]c^2$

$= [126(1.008665\ \mathrm{u}) + 82(1.007825\ \mathrm{u}) - 207.976652\ \mathrm{u}](931.50\ \mathrm{MeV/u}) = 1636.4\ \mathrm{MeV}$

$B/A = (1636.4\ \mathrm{MeV})/208 = 7.868\ \mathrm{MeV\ per\ nucleon}$

(b) $B = [78(1.008665\ \mathrm{u}) + 55(1.007825\ \mathrm{u}) - 132.905452\ \mathrm{u}](931.50\ \mathrm{MeV/u}) = 1118.5\ \mathrm{MeV}$

$B/A = (1118.5\ \mathrm{MeV})/133 = 8.410\ \mathrm{MeV\ per\ nucleon}$

(c) $B = [50(1.008665\ \mathrm{u}) + 40(1.007825\ \mathrm{u}) - 89.904704\ \mathrm{u}](931.50\ \mathrm{MeV/u}) = 783.9\ \mathrm{MeV}$

$B/A = (783.9\ \mathrm{MeV})/90 = 8.710\ \mathrm{MeV\ per\ nucleon}$

(d) $B = [32(1.008665\ \mathrm{u}) + 27(1.007825\ \mathrm{u}) - 58.933195\ \mathrm{u}](931.50\ \mathrm{MeV/u}) = 517.3\ \mathrm{MeV}$

$B/A = (517.3\ \mathrm{MeV})/59 = 8.768\ \mathrm{MeV\ per\ nucleon}$

RADIOACTIVE DECAY

2. A certain sample of a radioactive material decays at a rate of 548 per second at $t = 0$. At $t = 48$ minutes, the counting rate has fallen to 213 per second.
   a) What is the half-life of the radioactivity?
   b) What is its decay constant?
   c) What will be the decay rate at $t = 125$ minutes?

(a) With $a_0 = 548\ \text{s}^{-1}$ and $a = 213\ \text{s}^{-1}$ at $t = 48$ min, the radioactive decay equation $a = a_0 e^{-\lambda t}$ gives $213\ \text{s}^{-1} = (548\ \text{s}^{-1}) e^{-0.693(48\ \text{min})/t_{1/2}}$ or

$$\ln \frac{213\ \text{s}^{-1}}{548\ \text{s}^{-1}} = -\frac{0.693(48\ \text{min})}{t_{1/2}} \qquad \text{so} \qquad t_{1/2} = 35\ \text{min}$$

(b) $$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{35\ \text{min}} = 0.020\ \text{min}^{-1}$$

(c) $$a = a_0 e^{-\lambda t} = (548\ \text{s}^{-1}) e^{-(0.020\ \text{min}^{-1})(125\ \text{min})} = 46\ \text{s}^{-1}$$

3. Ordinary potassium contains 0.012 percent of the naturally occurring radioactive isotope ^{40}K, which has a half-life of $1.3 \cdot 10^9$ y.
   a) What is the activity of 1.0 kg of potassium?
   b) What would have been the fraction of ^{40}K in natural potassium $4.5 \cdot 10^9$ y ago?

(a) $$N = \frac{1000\ \text{g}}{39.1\ \text{g/mole}} 6.022 \times 10^{23}\ \text{atoms/mole} = 1.54 \times 10^{25}\ \text{atoms}$$

Of these, 0.012% or 1.85×10^{21} are radioactive ^{40}K. The activity is

$$a = \lambda N = \frac{0.693}{t_{1/2}} N = \frac{0.693}{(1.3 \times 10^9\ \text{y})(3.156 \times 10^7\ \text{s/y})} 1.85 \times 10^{21} = 3.1 \times 10^4\ \text{s}^{-1} = 0.85\ \mu\text{Ci}$$

(b) In a sample of N atoms, N_r are radioactive and N_nr are nonradioactive ($N = N_\text{r} + N_\text{nr}$). Presently, $N_\text{r}/N = 0.012\%$. With $N_\text{r} = N_{\text{r}0} e^{-\lambda t}$, we obtain for the original number of radioactive nuclei

$$N_{\text{r}0} = N_\text{r} e^{\lambda t} = N_\text{r} e^{0.693(4.5 \times 10^9\ \text{y})/(1.3 \times 10^9\ \text{y})} = 11.0 N_\text{r}$$

At that time (4.5×10^9 years ago), the total number was $N_0 = N_\text{nr} + N_{\text{r}0}$ and the fraction of ^{40}K was

$$\frac{N_{\text{r}0}}{N_0} = \frac{N_{\text{r}0}}{N_\text{nr} + N_{\text{r}0}} = \frac{11 N_\text{r}}{N_\text{nr} + 11 N_\text{r}} = \frac{11 N_\text{r}}{N + 10 N_\text{r}} = \frac{11(N_\text{r}/N)}{1 + 10(N_\text{r}/N)} = \frac{11(1.2 \times 10^{-4})}{1 + 10(1.2 \times 10^{-4})} = 0.13\%$$

DECAY PROCESSES

4. The natural decay chain $^{238}_{92}\mathrm{U} \rightarrow {}^{206}_{82}\mathrm{Pb}$ consists of several alpha decays and negative beta decays.
   a) How many alpha decays must occur in the chain?
   b) For that number of alpha decays, how many beta decays must occur to make the final Z come out correctly?
   c) Sometimes a decay chain can proceed through different branches and thus have different intermediate radionuclides. Do your answers to a) and b) depend on this branching?

(a) Mass number can only be made correct by alpha decays; beta decays do not influence the mass number. The mass number needs to decrease from 238 to 206 which is a decrease of 32. Each alpha decay reduces the mass number by 4. So 8 alpha decay events are needed.

(b) With 8 alpha decays, the atomic number Z will have decreased by 16, so $92 - 16 = 76$. It should become 82. Each beta decay event increases Z by 1. So 6 beta decay events are needed.

(c) No. The exact order of alpha and beta decays may change but in total the same number of alpha and beta events is needed.

5. Using the table below, show that $^{8}_{4}\mathrm{Be}$ can decay to two α-particles, but $^{12}_{6}\mathrm{C}$ cannot decay to three α-particles. How much energy is released at decay?

Nucleus	Binding Energy (MeV)	Binding energy per nucleon (MeV/A)
$^{2}_{1}\mathrm{H}$	2.22	1.1
$^{4}_{2}\mathrm{He}$	28.30	7.1
$^{8}_{4}\mathrm{Be}$	56.50	7.1
$^{12}_{6}\mathrm{C}$	92.16	7.7

An alpha particle is a helium nucleus ($^{4}_{2}\mathrm{He}$). We calculate the Q values for both proposed Beryllium (Q_b) and Carbon (Q_c) reactions:

$$Q_b = 2 \cdot 28.3 - 56.5 \text{ MeV} = +0.1 MeV$$
$$Q_c = 3 \cdot 28.3 - 92.16 \text{ MeV} = -7.26 MeV$$

Only reactions with a positive Q value are possible, so we have shown that beryllium can decay twice through Alpha radiation but carbon cannot decay three times.

6. Complete the following decays and compute the corresponding Q-values:

   a) $^{27}\mathrm{Si} \rightarrow {}^{27}\mathrm{Al} + \cdots$
   b) $^{93}\mathrm{Mo} + e^- \rightarrow \cdots$

(a) $^{27}_{14}\mathrm{Si} \rightarrow {}^{27}_{13}\mathrm{Al} + e^+ + \nu \quad Q : (26.9867 - 26.9867 - 26.9815 - 2 \cdot 0.000549) \cdot 931.05 = 3.8 \text{ MeV}$

(b) $^{93}_{42}\mathrm{Mo} + e^- \rightarrow {}^{93}_{41}\mathrm{Nb} + \nu \quad Q : (92.906813 - 92.90637) \cdot 931.05 = 0.41 \text{ MeV}$

7. The atomic mass of $^{211}_{84}$Po is 210.986627 u, and it decays by α-emission to $^{208}_{82}$Pb, which has an atomic mass of 206.975872 u.
   a) Calculate the decay energy Q and compare this to the rest energy of the least massive particle. Can we reasonably conclude that we may ignore relativistic effects in this fission reaction? *(Note: Your answer here holds in general for α-emissions, even though you only analyze it in this case)*
   b) Assuming that the parent has no kinetic energy before the reaction and that all of the decay energy is distributed between the α-particle and daughter nucleus, derive the formula that gives the distribution of Q between α-particle and daughter.
   c) Calculate the actual kinetic energies of the α-particle and daughter nucleus. Compare the kinetic energy of the α-particle with the value on the chart of nuclei.

(a) $Q = \left(m(^{211}_{84}\text{Po}) - m(^{207}_{82}\text{Pb}) - m(^{4}_{2}\text{He})\right)c^2 = (210.986627 - 206.975872 - 4.002602) \cdot 931.5 = 7.59 \text{ MeV}.$

$m(^{4}_{2}\text{He}) = 4.002602 \cdot 931.5 = 3.728423763 \text{ TeV} \Rightarrow \dfrac{Q}{m(^{4}_{2}\text{He})} \approx 0.00203.$

The Restmass of the Helium nucleus is about 500 times larger than the maximal kinetic energy said nucleus. We may therefore disregard relativistic effects.

(b) First some notation: We have to use equations for momentum and kinetic energy. As we have analyzed in part a) we may disregard relativistic effects. This means we have the following equations, where the subscript x is the daughter nucleus:

$$Q = E_k = K_\alpha + K_x \tag{1}$$
$$0 = p_\alpha + p_x \underbrace{\Longrightarrow}_{\text{Abs vals}} |p_\alpha| = |p_x| \tag{2}$$

Now we use the classical relation between momentum and kinetic energies and solve one of the energies in terms of the other and their masses:

$$|p_\alpha| = |p_x| \Rightarrow m_\alpha v_\alpha = m_x v_x \Rightarrow (m_\alpha v_\alpha)^2 = (m_x v_x)^2 \Rightarrow 2m_\alpha \left(\frac{1}{2} m_\alpha v_\alpha^2\right) = 2m_x \left(\frac{1}{2} m_x v_x^2\right) \Rightarrow$$
$$2m_\alpha K_\alpha = 2m_x K_x \Longrightarrow K_x = \frac{m_\alpha}{m_x} K_\alpha$$

As expected, we plug in the values into equation (1):

$$Q = K_\alpha + K_x = K_\alpha + \frac{m_\alpha}{m_x} K_\alpha = \frac{m_x + m_\alpha}{m_x} K_\alpha \Longrightarrow K_\alpha = \frac{m_x}{m_x + m_\alpha} Q$$

Now we approximate $m_x \approx (A-4)u$, $m_\alpha \approx 4u$ (Why is this reasonable?) to finally get the desired expression:

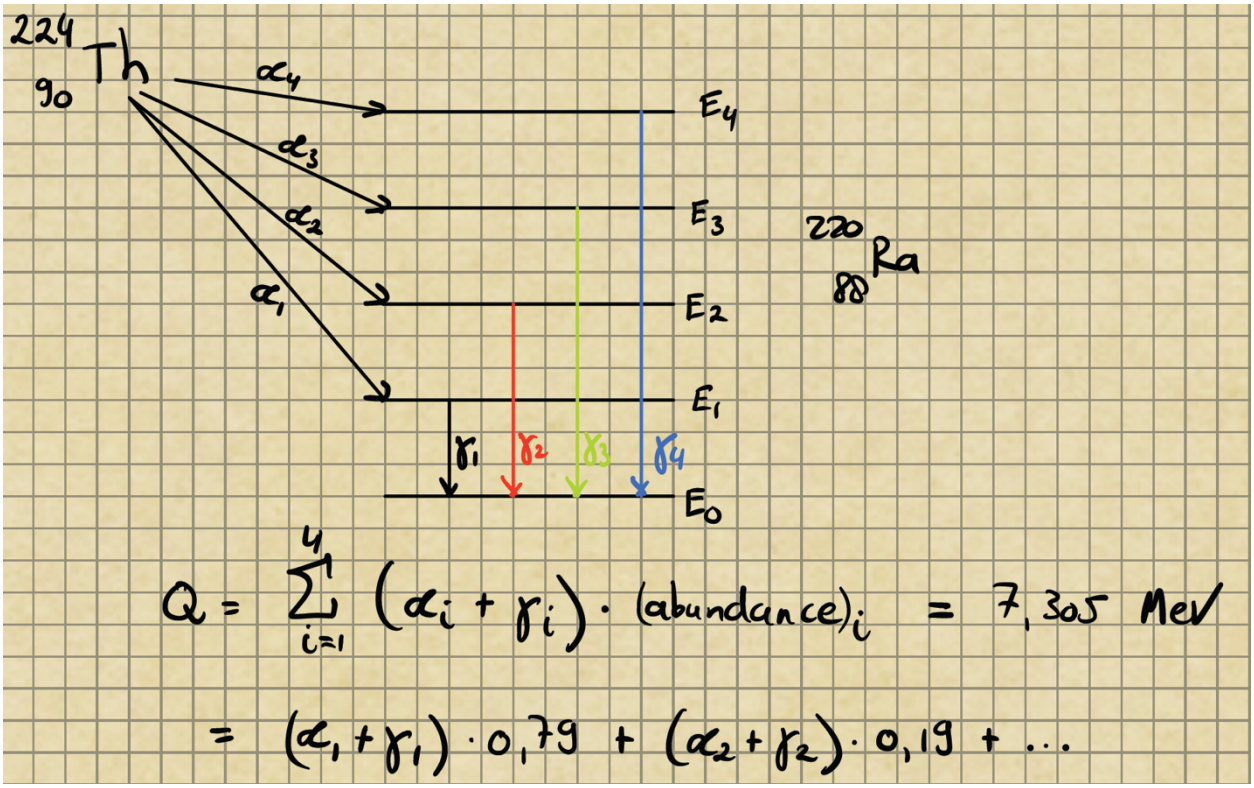
$$\frac{m_x}{m_x + m_\alpha} Q \approx \frac{(A-4)u}{(A-4)u + 4u} Q \Longrightarrow K_\alpha \approx \frac{A-4}{A} Q$$

(c) Simply plug in the values to obtain:

$$K_\alpha \approx \left(\frac{207 - 4}{207}\right) \cdot 7.59 = 7.44 \text{MeV}$$
$$K_x = Q - K_\alpha \approx 7.59 - 7.44 = 0.15 \text{MeV}$$

8. Construct a decay diagram for $^{224}_{90}\mathrm{Th}$, which decays by α-emission to $^{220}_{88}\mathrm{Ra}$. The decay energy Q is 7.305 MeV, the half life is 1.045 s and the energies of emissions are given as:

Emission	Energy (MeV)	Abundance (%)
α_1	7.170	79
α_2	6.997	19
α_3	6.768	1.2
α_4	6.702	0.6
γ_1	0.177	
γ_2	0.235	
γ_3	0.297	
γ_4	0.410	



9. The fission of uranium-235 can occur in several ways. One possible fission reaction is $^{235}\mathrm{U} + n \rightarrow {}^{141}\mathrm{Ba} + {}^{92}\mathrm{Kr} + 3n$ where n is a neutron. The reaction produces 3 neutrons which may give rise to new fission reactions.
   a) Compute the energy that is produced ("Q value") by one fission reaction of this type.
   b) Compute the total amount of energy in Joules if all atoms in 235 gram of $^{235}\mathrm{U}$ perform this reaction.
   c) The heat of combustion of methane ($\mathrm{CH_4}$) is about 889 kJ/mol. How much energy (in joules) is produced if 235 grams of methane are combusted?

(a) $Q = \Big(m(^{235}_{92}\mathrm{U}) + m(n) - m(^{141}_{56}\mathrm{Ba}) - m(^{92}_{36}\mathrm{Kr}) - 3m(n)\Big)$
$= \Big(235,0439299 - 140,914411 - 91,926156 - 2 \cdot 1.00867\Big) \cdot 931.5 = 173.28 \text{ MeV}$

(b) Number of atoms in 235 g $^{235}_{92}\mathrm{U}$: approximately $N_A = 6.02 \cdot 10^{23}$
Total energy produced: $E = 6.02 \cdot 10^{23} \cdot 173.28 \cdot 10^{6} \cdot 1.602 \cdot 10^{-19} = 1.67 \cdot 10^{13}$ J

(c) Number of mol in 235 g $\mathrm{CH_4}$: $\frac{235}{16} = 14.69$ mol.
Total energy if 235 g methane is combusted: $E = 14.69 \cdot 889 = 1.300 \cdot 10^{4}$ J